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Advanced Synthesis & Catalysis

Iron(III) Chloride-Catalyzed Tandem Aza-Prins/Friedel–Crafts Cyclization of 2-Arylethyl-2,3-butadienyl Tosylamides and Aldehydes-An Efficient Synthesis of Benzo[*f*]isoquinolines

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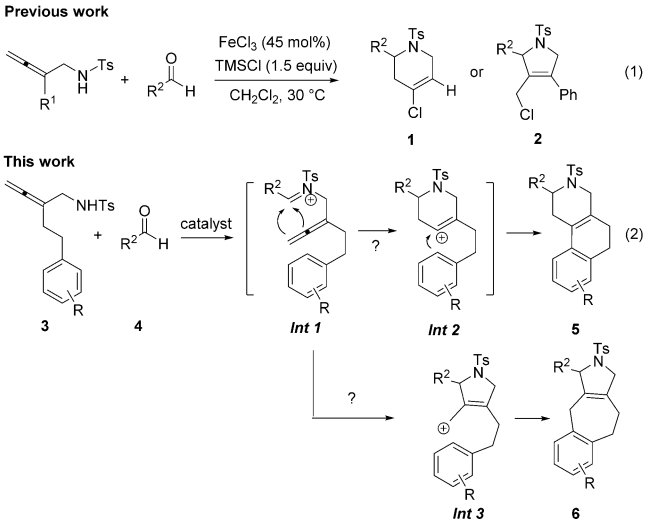
Supporting information for this article is available on the WWW under http://dx.doi.org/10.1002/adsc.201600107.

Abstract: An efficient approach for the synthesis of 1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinolines *via* a tandem aza-Prins/Friedel–Crafts cyclization from 2-arylethyl-2,3-butadienyl tosylamides and aldehydes has been developed. This iron(III) chloride-catalyzed cascade cyclization at the room temperature with different types of aldehydes, such as aromatic and heteroaromatic aldehydes, alkyl aldehydes, and α,β-unsaturated aldehydes, affords the products in moderate to excellent yields (up to 97%). In this reaction, chlorotrimethylsilane was applied to activate the aldehydes. The cheap catalyst, the mild reaction conditions, and the broad scope of the substrates make this method highly useful.

Keywords: allene; aza-Prins cyclization; benzo[*f*]isoquinoline; cascade; iron(III) chloride

Introduction

Benzo[*f*]isoquinoline alkaloids are a class of important heterocycles with interesting pharmacological activities,[1,2] such as high-affinity σ ligands serving as potentially therapeutic agents for the treatment of Parkinson's disease and schizophrenia[1b–f] and potential calcium channel blockers.[2] Thus, much attention has been paid to the synthesis of this class of compounds.[1e,3–7] Prins cyclization using alkenes and alkynes has became a powerful synthetic method for the synthesis of heterocyclic compounds.[8,9] However, Prins cyclization utilizing allenes is still not well explored for the issue of regioselectivity.[10] In 2012, we reported that 2,3-butadienyl tosylamides with aldehydes produced 2,5-dihydro-1*H*-pyrrole or 1,2,3,6-tetrahydropyridine derivatives respectively, depending on the nature of the R^1 substituent. (Scheme 1, eq. 1).[10c] We envisioned that when an aryl group is tethered into the R^1 substituent with two carbon atom chain, 2-arylethyl-2,3-butadienyl tosylamides would react with aldehydes to provide iminiums ***Int-1*** (Scheme 1, eq. 2). We wondered whether sequential Friedel–Crafts type cyclizations would provide tricyclic products **5** or **6** either *via* ***Int-2*** or ***Int-3*** or a concerted process. Herein, we report our recent observations on this topic.



Scheme 1. Aza-Prins reaction of 2,3-butadienyl tosylamides.

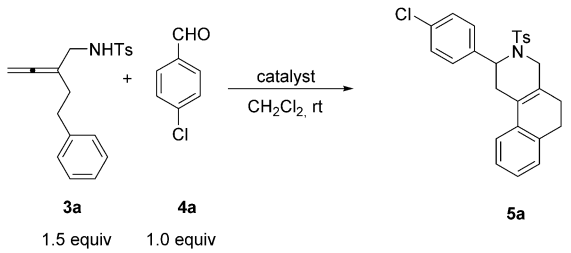
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Results and Discussion

We began our study with 2-(2-phenylethyl)-2,3-butadienyl tosylamide **3a**, which could be easily obtained based on a known method, developed by us,[11] and *p*-chlorobenzaldehyde **4a**. To our delight, under the catalysis of $FeCl_3$ (20 mol%) in CH_2Cl_2 with stirring at room temperature for 52 h, one of the two expected products, hexahydrobenzo[*f*]isoquinoline **5a**, was obtained in 43% yield (Table 1, entry 1).[12] The structure of this product was confirmed by its X-ray crystallographic study (Figure 1).[13] The formation of **6a** *via* ***Int-3***-type intermediate was not observed.

Increasing the amount of the catalyst did not improve the yield significantly (Table 1, entries 2–3).

Table 1. Optimization of Lewis acid catalyzed cascade aza-Prins cyclization of 2-phenylethyl-2,3-butadienyl tosylamide **3a**.[a]



Entry	Cat. (mol%)	TMSCl (equiv)	*t* (h)	Yield of **5a** (%)[b]	Recovery of **4a** (%)[b]
1	$FeCl_3$ (20)	–	52	43	4
2	$FeCl_3$ (30)	–	27	51	30
3	$FeCl_3$ (60)	–	13	65	3
4	$In(OTf)_3$ (10)	–	10.5	–	100
5	$Sc(OTf)_3$ (30)	–	48	–	100
6	$Cu(OTf)_2$ (30)	–	48	–	100
7	$Cu(acac)_2$ (30)	–	48	–	100
8	$FeCl_3$ (20)	2.5	20.5	83	0
9	$FeCl_3$ (20)	1.5	23	89 (81[c])	0
10	$FeCl_3$ (20)	1.0	23	85	5
11	$FeCl_3$ (20)	0.5	23	82	7
12	$FeCl_3$ (20)	0.2	24	60	28
13[d]	$FeCl_3$ (20)	1.5	23	85	8
14[e]	$FeCl_3$ (20)	1.5	24	51	17

[a] The reaction was conducted using **3a** (1.5 equiv), aldehyde ($c = 0.1$ M), catalyst and TMSCl in CH_2Cl_2 at room temperature.
[b] Determined by ^{1}H NMR analysis with 1,3,5-trimethylbenzene as the internal standard.
[c] Isolated yields and purity (95%) determined by using CH_2Br_2 as the internal standard.
[d] 1.2 equiv of **3a** was used in this reaction.
[e] 1.0 equiv of **3a** was used in this reaction.

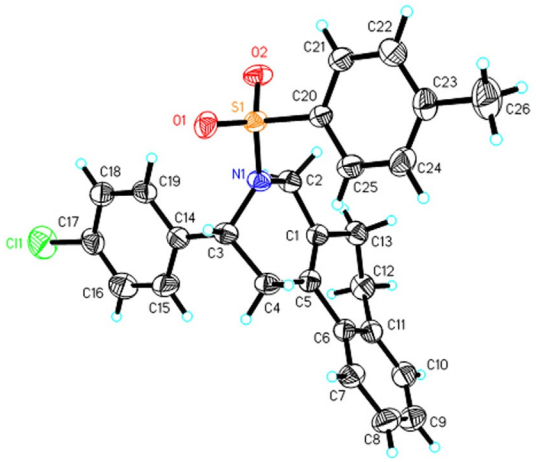


Figure 1. ORTEP representation of compound **5a**

The desired product **5a** was not formed under other metal catalysts such as $In(OTf)_3$, $Sc(OTf)_3$, $Cu(OTf)_2$, and $Cu(acac)_2$ (Table 1, entries 4–7). The yield of the product **5a** improved greatly to 83% by the addition of 2.5 equiv of TMSCl (Table 1, entry 8), which may activate the carbonyl group.[12b,14] Furthermore, when the amount of TMSCl was reduced to 1.5 equiv, the product **5a** was obtained in 89%. This was found to be the optimum amount of this additive (Table 1, entries 9–12). In addition, the yield of the reaction dropped when the amount of **3a** was reduced (Table 1, entries 13–14). Thus, **3a** (1.5 equiv), **4a** (1.0 equiv), $FeCl_3$ (20 mol%), and TMSCl (1.5 equiv) in CH_2Cl_2 at room temperature were defined as the optimized reaction conditions for further study.

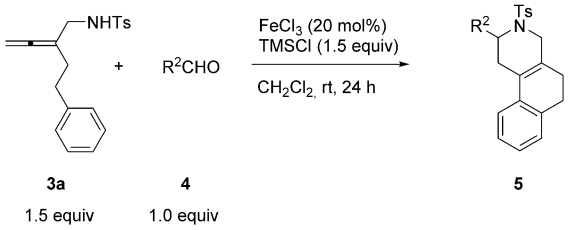
To examine the scope of the cascade aza-Prins/Friedel–Crafts cyclization, diversified aldehydes were examined for their reaction with **3a** under the optimal reaction conditions. In addition to the *p*-chlorobenzaldehyde **4a**, benzaldehyde with different substituents, such as electron-donating *p*-MeO and electronic-withdrawing *p*-F, *p*-Br, *p*-I, as well as *o*-Cl groups, could all afford the desired products **5** in good to excellent yields (Table 2, entries 1–6). 2-Naphthaldehyde also reacted smoothly to afford the desired product **5h** in an excellent yield (Table 2, entry 7). The scope of this reaction may also be extended to heteroaromatic aldehydes, such as 3-thiophenaldehyde, 3-furaldehyde and *N*-Ts-indole-3-carboxaldehyde (Table 2, entries 8–10). Cyclic and aliphatic aldehydes with Cl or phenyl substituents, and even the paraformaldehyde afforded the desired products **5l–5o** successfully (Table 2, entries 11–14).

Excitingly, α,β-unsaturated alk-2-ynals, such as 2-nonynal **4p** and 2-propynal **4q** also afforded the corresponding products (Scheme 2). These types of functionalized products provide opportunities for further synthetic elaboration.

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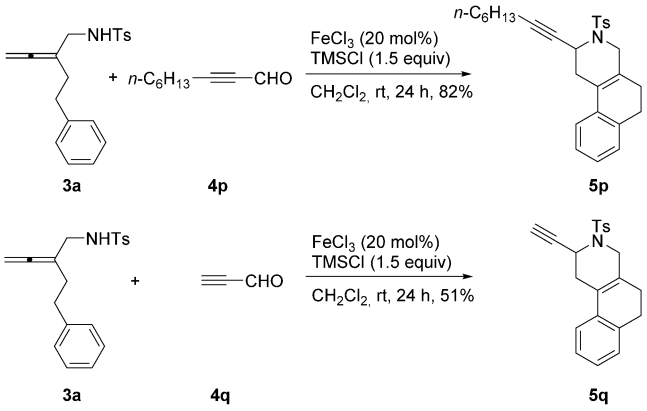
16154169, 2016, 12, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201600107 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. The scope of $FeCl_3$-catalyzed cascade aza-Prins cyclization of 2-phenylethyl-2,3-butadienyl tosylamide.[a]



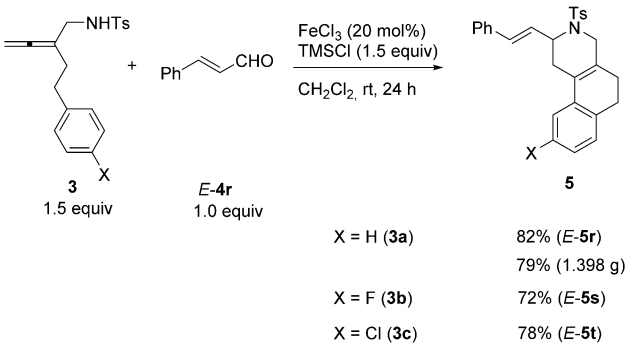
Entry	R^2	Yield of **5** (%)[b]
1	*p*-BrC_6H_4 (**4b**)	88 (**5b**)
2	*p*-IC_6H_4 (**4c**)	88 (**5c**)
3	*p*-FC_6H_4 (**4d**)	85 (**5d**)
4[c]	*p*-$MeOC_6H_4$ (**4e**)	83 (**5e**)
5	*o*-ClC_6H_4 (**4f**)	67 (**5f**)
6	Ph (**4g**)	79 (**5g**)
7	2-naphthyl (**4h**)	84 (**5h**)
8	3-thienyl (**4i**)	83 (**5i**)
9	3-furyl (**4j**)	51 (**5j**)
10	*N*-Ts-indol-3-yl (**4k**)	64 (**5k**)
11	$Cl(CH_2)_4$ (**4l**)	97 (**5l**)
12	Cy (**4m**)	70 (**5m**)
13	$PhCH_2CH_2$ (**4n**)	75 (**5n**)
14	H (**4o**)	78 (**5o**)

[a] The reaction was carried out at room temperature in CH_2Cl_2 using aldehyde ($c = 0.1$ M), **3** (1.5 equiv), $FeCl_3$ (20 mol%), TMSCl (1.5 equiv) at indicated time.
[b] The yields were isolated yields.
[c] $FeCl_3$ (30 mol%) was used.



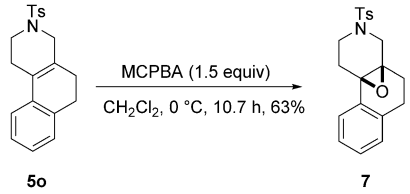
Scheme 2. Cascade aza-Prins cyclization with α,β-unsaturated aldehydes.

In addition, cinnamaldehyde also afforded the desired product *E*-**5r** in excellent yield. This reaction was readily conducted on a 1.4 g scale. Besides, the reactions of 2-arylethyl-2,3-butadienyl tosylamides with different substituents, such as *p*-F or *p*-Cl groups to the aryl group also afforded the corresponding hexahydrobenzo[*f*]isoquinolines *E*-**5s** and *E*-**5t** in good yields (Scheme 3).



Scheme 3. Cascade aza-Prins cyclization of 2-arylethyl-2,3-butadienyl tosylamides with a *p*-substituent on the aryl group.

For the purpose of showing the synthetic potential of our compounds, epoxidation of the C=C bond of 1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline derivative **5o** was realized with 3-chloroperoxybenzoic acid as oxidant (Scheme 4).[15]



Scheme 4. Epoxidation of 1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline derivative **5o**

Conclusions

In conclusion, we have developed an efficient methodology for the synthesis of 1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinolines *via* novel $FeCl_3$ catalysed aza-Prins-intramolecular Friedel–Crafts cascade cyclization from readily available 2-arylethyl-2,3-butadienyl tosylamides and different types of aldehydes. The mild reaction conditions (rt), the cheap catalyst ($FeCl_3$), the broad scope of the substrates with the potential manipulation of the installed functionality, and the efficiency to the pharmacologically important benzo[*f*]isoquinolines skeletons make this method highly interesting to the organic chemists and pharmacologists.

Experimental Section

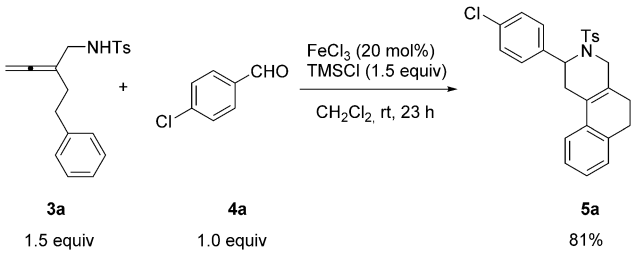
General Information: All reactions were carried out in oven dried Schlenk tubes. DCM (dichloromethane) were purchased from Sinopharm and used directly without further treatment. All the temperatures are referred to the oil baths used. $FeCl_3$ was purchased from Sinopharm. All ^1H NMR

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experiments were measured with tetramethylsilane (0 ppm) as the internal reference in $CDCl_3$ and ^{13}C NMR experiments were measured in relative to the signal of $CDCl_3$ (77.0 ppm). ^{19}F NMR experiments were measured with Cl_3CF (0 ppm) as the internal reference.

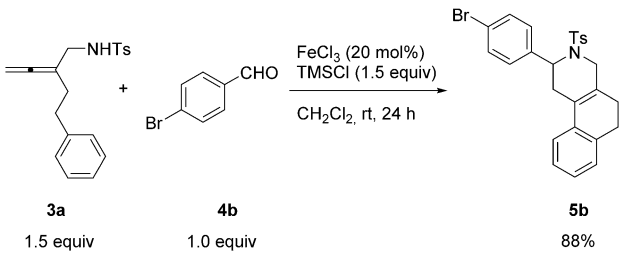
1. Preparation of 2-(4-chlorophenyl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5a



Typical Procedure I: To a dried Schlenk tube were added $FeCl_3$ (6.5 mg, 0.040 mmol)/CH_2Cl_2 (0.5 mL), **3a** (98.1 mg, 0.30 mmol)/CH_2Cl_2 (0.5 mL), **4a** (28.2 mg, 0.20 mmol)/CH_2Cl_2 (0.5 mL), and TMSCl (33.0 mg, 0.30 mmol)/CH_2Cl_2 (0.5 mL) sequentially. The mixture was then stirred at room temperature until the completion of the reaction as monitored by TLC (23 h). The resulting mixture was diluted with CH_2Cl_2 (5 mL) and diethyl ether (10 mL). Then anhydrous $MgSO_4$ was added and the resulting mixture was then filtered through a short pad of silica gel to remove the inorganic salts; eluent: 6×[CH_2Cl_2 (5 mL) + Et_2O (10 mL)]. After evaporation, the mixture was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 20:1:1) to afford **5a** (76.7 mg, 81%, purity = 95% based on the ^1H NMR analysis with CH_2Br_2 as the internal standard) as a solid; m.p. 154–156 °C (petroleum ether/CH_2Cl_2); ^1H NMR (400 MHz, $CDCl_3$) δ 7.68 (d, J = 8.4 Hz, 2H, ArH), 7.30–7.08 (m, 9H, ArH), 7.05 (d, J = 7.2 Hz, 1H, ArH), 5.43 (d, J = 6.0 Hz, 1H, TsNCH), 4.08 (d, J = 18.8 Hz, 1H, one proton of CH_2), 3.48 (d, J = 18.8 Hz, 1H, one proton of CH_2), 2.85–2.50 (m, 4H, 2×CH_2), 2.35 (s, 3H, CH_3 of Ts), 2.20–2.07 (m, 1H, one proton of CH_2), 2.02–1.92 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.4, 137.44, 137.37, 135.1, 134.2, 133.3, 129.6, 129.5, 128.9, 128.5, 127.5, 126.9, 126.8, 126.5, 123.9, 121.4, 52.7, 44.0, 27.6, 25.70, 25.69, 21.4; MS (EI) *m/z* (%) 451 (M^+(^{37}Cl), 6.28), 449 (M^+(^{35}Cl), 13.80), 293 (100); IR (neat) 3070, 3018, 2924, 2876, 2839, 1591, 1491, 1443, 1340, 1261, 1161, 1095, 1052, 1011 cm^{-1}; anal calcd for $C_{26}H_{24}NO_2SCl$: C, 69.40; H, 5.38; N, 3.11; found: C, 69.27; H, 5.44; N, 3.11.

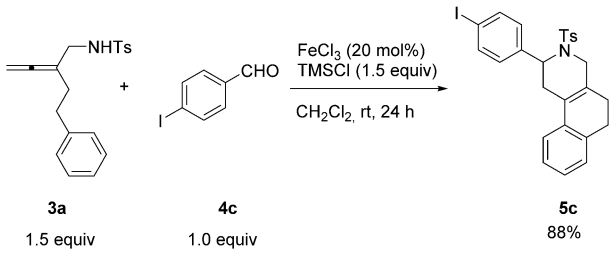
2. Preparation of 2-(4-bromophenyl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5b

Following **Typical Procedure I**, the reaction of $FeCl_3$ (6.5 mg, 0.040 mmol), **3a** (96.4 mg, 0.30 mmol), **4b** (36.7 mg, 0.20 mmol), TMSCl (32.6 mg, 0.30 mmol), and CH_2Cl_2 (2 mL) afford the desired product **5b** (86.0 mg, 88%) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 20:1:1) as



a solid; m.p. 111–112 °C (petroleum ether/CH_2Cl_2); ^1H NMR (400 MHz, $CDCl_3$) δ 7.68 (d, J = 8.0 Hz, 2H, ArH), 7.35 (d, J = 8.8 Hz, 2H, ArH), 7.23–7.09 (m, 7H, ArH), 7.05 (d, J = 7.2 Hz, 1H, ArH), 5.41 (d, J = 6.0 Hz, 1H, TsNCH), 4.08 (d, J = 18.8 Hz, 1H, one proton of CH_2), 3.48 (d, J = 18.8 Hz, 1H, one proton of CH_2), 2.80–2.56 (m, 4H, 2×CH_2), 2.35 (s, 3H, CH_3 of Ts), 2.20–2.07 (m, 1H, one proton of CH_2), 2.02–1.92 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.4, 138.0, 137.3, 135.1, 134.1, 131.4, 129.6, 129.5, 129.2, 127.4, 126.9, 126.8, 126.5, 123.9, 121.43, 121.42, 52.8, 44.0, 27.6, 25.7, 25.6, 21.4; MS (EI) *m/z* (%) 495 (M^+(^{81}Br), 3.41), 493 (M^+(^{79}Br), 2.99), 91 (100); IR (neat) 2924, 2853, 1597, 1488, 1445, 1340, 1264, 1159, 1095, 1075, 1051, 1008 cm^{-1}; anal calcd for $C_{26}H_{24}NO_2SBr$: C, 63.16; H, 4.89; N, 2.83; found: C, 63.02; H, 4.81; N, 2.58.

3. Preparation of 2-(4-iodophenyl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5c

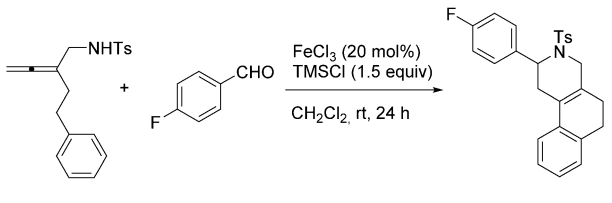


Following **Typical Procedure I**, the reaction of $FeCl_3$ (6.5 mg, 0.040 mmol), **3a** (98.0 mg, 0.30 mmol), **4c** (46.0 mg, 0.20 mmol), TMSCl (32.6 mg, 0.30 mmol), and CH_2Cl_2 (2 mL) afford the desired product **5c** (94.8 mg, 88%) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 20:1:1) as a solid; m.p. 177–178 °C (petroleum ether/CH_2Cl_2); ^1H NMR (300 MHz, $CDCl_3$) δ 7.68 (d, J = 8.4 Hz, 2H, ArH), 7.55 (d, J = 8.7 Hz, 2H, ArH), 7.24–7.00 (m, 8H, ArH), 5.40 (d, J = 6.0 Hz, 1H, TsNCH), 4.09 (d, J = 18.9 Hz, 1H, one proton of CH_2), 3.48 (d, J = 19.2 Hz, 1H, one proton of CH_2), 2.80–2.50 (m, 4H, 2×CH_2), 2.36 (s, 3H, CH_3 of Ts), 2.18–1.91 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.4, 138.7, 137.4, 137.3, 135.1, 134.1, 129.6, 129.5, 129.4, 127.4, 126.9, 126.8, 126.5, 123.9, 121.4, 93.1, 52.9, 44.1, 27.6, 25.7, 25.6, 21.4; MS (EI) *m/z* (%) 541 (M^+, 3.86), 385 (100); IR (neat) 2962, 2925, 2854, 1485, 1444, 1338, 1260, 1158, 1093, 1006 cm^{-1}; HRMS (EI) calcd for $C_{26}H_{24}NO_2SI$ (M^+): 541.0573; found: 541.0573.

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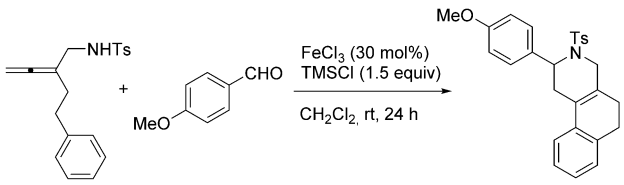
4. Preparation of 2-(4-fluorophenyl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 3d



3a 1.5 equiv | **4d** 1.0 equiv | **5d** 85%

Following **Typical Procedure I**, the reaction of $FeCl_3$ (6.6 mg, 0.040 mmol), **3a** (96.4 mg, 0.30 mmol), **4d** (25.4 mg, 0.20 mmol), TMSCl (32.7 mg, 0.30 mmol), and CH_2Cl_2 (2 mL) afford the desired product **5d** (75.6 mg, 85%) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 20:1:1) as a solid; m.p. 67–68 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.68 (d, J = 8.0 Hz, 2H, ArH), 7.28 (dd, J = 8.6, 5.4 Hz, 2H, ArH), 7.23–7.09 (m, 5H, ArH), 7.06 (d, J = 7.6 Hz, 1H, ArH), 6.92 (t, J = 8.8 Hz, 2H, ArH), 5.45 (d, J = 6.0 Hz, 1H, TsNCH), 4.09 (d, J = 18.8 Hz, 1H, one proton of CH_2), 3.48 (d, J = 18.8 Hz, 1H, one proton of CH_2), 2.80–2.58 (m, 4H, 2 × CH_2), 2.35 (s, 3H, CH_3 of Ts), 2.21–2.08 (m, 1H, one proton of CH_2), 2.05–1.94 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 162.0 (d, J = 244.8 Hz), 143.3, 137.4, 135.1, 134.7 (d, J = 3.4 Hz), 134.2, 129.6, 129.5, 129.2 (d, J = 8.0 Hz), 127.4, 126.84, 126.78, 126.5, 123.9, 121.4, 115.1 (d, J = 21.3 Hz), 52.6, 44.0, 27.6, 25.9, 25.7, 21.4; ^{19}F NMR (376 MHz, $CDCl_3$) −115.4; MS (EI) *m/z* (%) 433 (M^+, 8.77), 277 (100); IR (neat) 2962, 2925, 2854, 1601, 1509, 1445, 1338, 1306, 1261, 1226, 1158, 1093, 1016 cm^{-1}; HRMS (EI) calcd for $C_{26}H_{24}NO_2SF$ (M^+): 433.1512; found: 433.1509.

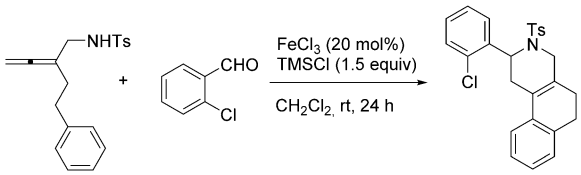
5. Preparation of 2-(4-methoxyphenyl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5e



3a 1.5 equiv | **4e** 1.0 equiv | **5e** 83%

Following **Typical Procedure I**, the reaction of $FeCl_3$ (9.6 mg, 0.060 mmol), **3a** (97.4 mg, 0.30 mmol), **4e** (27.8 mg, 0.20 mmol), TMSCl (32.6 mg, 0.30 mmol), and CH_2Cl_2 (2 mL) afford the desired product **5e** (75.3 mg, 83%) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 30:1:1) as an oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.68 (d, J = 8.0 Hz, 2H, ArH), 7.26–7.08 (m, 7H, ArH), 7.06 (d, J = 7.2 Hz, 1H, ArH), 6.76 (d, J = 8.8 Hz, 2H, ArH), 5.44 (d, J = 6.2 Hz, 1H, TsNCH), 4.07 (d, J = 18.8 Hz, 1H, one proton of CH_2), 3.75 (s, 3H, CH_3), 3.49 (d, J = 18.8 Hz, 1H, one proton of CH_2), 2.79–2.56 (m, 4H, 2 × CH_2), 2.34 (s, 3H, CH_3 of Ts), 2.21–2.07 (m, 1H, one proton of CH_2), 2.04–1.93 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 158.8, 143.1, 137.6, 135.2, 134.4, 130.9, 129.51, 129.48, 128.7, 127.4, 126.8, 126.7, 126.5, 124.2, 121.5, 113.6, 55.1, 52.8, 43.9, 27.6, 25.9, 25.7, 21.4; MS (EI) *m/z* (%) 445 (M^+, 3.51), 289 (100); IR (neat) 2930, 2835, 1610, 1583, 1490, 1442, 1338, 1304, 1251, 1180, 1162, 1095, 1035, 1008 cm^{-1}; HRMS (EI) calcd for $C_{27}H_{27}NO_3S$ (M^+): 445.1712; found: 445.1715.

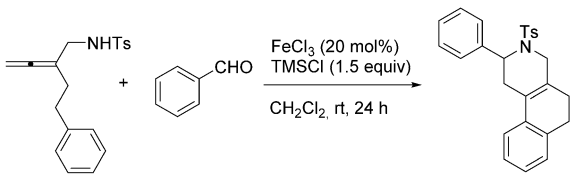
6. Preparation of 2-(2-chlorophenyl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5f



3a 1.5 equiv | **4f** 1.0 equiv | **5f** 67%

Following **Typical Procedure I**, the reaction of $FeCl_3$ (6.5 mg, 0.040 mmol), **3a** (97.4 mg, 0.30 mmol), **4f** (28.0 mg, 0.20 mmol), TMSCl (32.6 mg, 0.30 mmol), and CH_2Cl_2 (2 mL) afford the desired product **5f** (60.3 mg, 67%) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 20:1:1) as a solid; m.p. 73–75 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.65 (d, J = 8.0 Hz, 2H, ArH), 7.32 (d, J = 7.6 Hz, 1H, ArH), 7.18–7.05 (m, 7H, ArH), 7.03–6.93 (m, 2H, ArH), 5.88 (d, J = 6.0 Hz, 1H, TsNCH), 4.14 (d, J = 18.0 Hz, 1H, one proton of CH_2), 3.79 (d, J = 18.4 Hz, 1H, one proton of CH_2), 2.98–2.86 (m, 1H, one proton of CH_2), 2.76–2.65 (m, 3H, CH_2 and one proton of CH_2), 2.29 (s, 3H, CH_3 of Ts), 2.23–2.00 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.2, 138.2, 136.3, 135.0, 134.0, 133.2, 129.9, 129.2, 129.1, 128.6, 127.8, 127.3, 127.2, 126.8, 126.5, 126.4, 124.3, 121.6, 51.3, 45.5, 28.1, 27.7, 26.1, 21.3; MS (EI) *m/z* (%) 91 (100); IR (neat) 2924, 2852, 1597, 1471, 1442, 1381, 1345, 1262, 1162, 1094, 1039, 1014 cm^{-1}; anal calcd for $C_{26}H_{24}NO_2SCl$: C, 69.40; H, 5.38; N, 3.11; found: C, 69.30; H, 5.48; N, 2.93.

7. Preparation of 2-phenyl-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5g



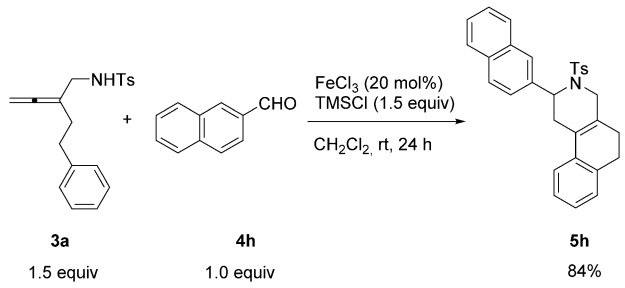
3a 1.5 equiv | **4g** 1.0 equiv | **5g** 79%

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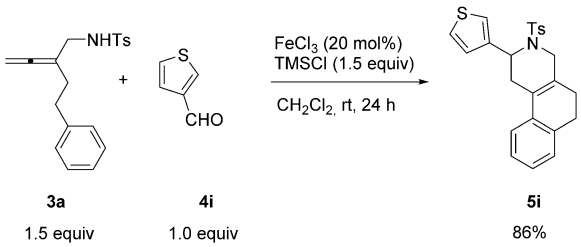
Following **Typical Procedure I**, the reaction of $FeCl_3$ (6.5 mg, 0.040 mmol), **3a** (96.2 mg, 0.30 mmol), **4g** (21.2 mg, 0.20 mmol), TMSCl (34.2 mg, 0.30 mmol), and CH_2Cl_2 (2 mL) afford the desired product **5g** (65.2 mg, 79%) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 20:1:1) as an oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.68 (d, J = 8.4 Hz, 2H, ArH), 7.35–7.27 (m, 2H, ArH), 7.26–7.02 (m, 9H, ArH), 5.47 (d, J = 6.4 Hz, 1H, TsNCH), 4.09 (d, J = 18.8 Hz, 1H, one proton of CH_2), 3.50 (d, J = 18.8 Hz, 1H, one proton of CH_2), 2.85–2.56 (m, 4H, 2×CH_2), 2.33 (s, 3H, CH_3 of Ts), 2.21–2.06 (m, 1H, one proton of CH_2), 2.04–1.93 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.2, 138.9, 137.5, 135.2, 134.3, 129.5, 129.4, 128.3, 127.43, 127.38, 126.8, 126.7, 126.5, 124.1, 121.5, 53.3, 44.1, 27.6, 25.73, 25.70, 21.4; MS (EI) m/z (%) 415 (M^+, 9.07), 259 (100); IR (neat) 2960, 2925, 2854, 1599, 1493, 1450, 1381, 1339, 1260, 1158, 1094, 1015 cm^{-1}; HRMS (EI) calcd for $C_{26}H_{25}NO_2S$ (M^+): 415.1606; found: 415.1603.

8. Preparation of 2-(2-naphthalenyl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*] isoquinoline 5h



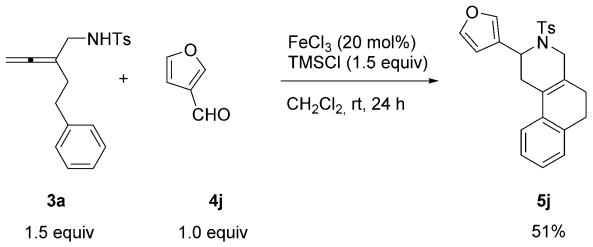
Following **Typical Procedure I**, the reaction of $FeCl_3$ (6.5 mg, 0.040 mmol), **3a** (98.2 mg, 0.30 mmol), **4h** (31.6 mg, 0.20 mmol), TMSCl (32.6 mg, 0.30 mmol), and CH_2Cl_2 (2 mL) afford the desired product **5h** (82.0 mg, 84%, purity = 96% based on the ^{1}H NMR analysis with CH_2Br_2 as the internal standard) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 20:1:1) as a solid; m.p. 95–96 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.81–7.68 (m, 4H, Ar-H), 7.68–7.58 (m, 2H, Ar-H), 7.52 (d, J = 8.8 Hz, 1H, ArH), 7.47–7.36 (m, 2H, Ar-H), 7.25–7.04 (m, 6H, Ar-H), 5.61 (d, J = 6.4 Hz, 1H, TsNCH), 4.12 (d, J = 18.8 Hz, 1H, one proton of CH_2), 3.50 (d, J = 18.8 Hz, 1H, one proton of CH_2), 2.91 (d, J = 17.2 Hz, 1H, one proton of CH_2), 2.75–2.63 (m, 3H, one proton of CH_2 + CH_2), 2.31 (s, 3H, CH_3 of Ts), 2.20–2.05 (m, 1H, one proton of CH_2), 1.99–1.88 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.2, 137.5, 136.2, 135.2, 134.4, 132.9, 132.6, 129.6, 129.5, 128.2, 128.0, 127.4, 126.8, 126.7, 126.5, 126.0, 125.94, 125.89, 125.8, 124.2, 121.5, 53.4, 44.2, 27.6, 25.7, 25.6, 21.4; MS (EI) m/z (%) 465 (M^+, 6.08), 141 (100); IR (neat) 2962, 2925, 1443, 1338, 1260, 1157, 1093, 1013 cm^{-1}; HRMS (EI) calcd for $C_{30}H_{27}NO_2S$ (M^+): 465.1763; found: 465.1760.

9. Preparation of 2-(3-thiophenyl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5i



Typical Procedure II: To a dried Schlenk tube were added $FeCl_3$ (32.0 mg, 0.20 mmol)/CH_2Cl_2 (2.5 mL), **3a** (491.2 mg, 1.50 mmol)/CH_2Cl_2 (2.5 mL), **4i** (113.4 mg, 1.00 mmol)/CH_2Cl_2 (2.5 mL), and TMSCl (168.0 mg, 1.50 mmol)/CH_2Cl_2 (2.5 mL) sequentially. The mixture was then stirred at room temperature until the completion of the reaction as monitored by TLC (24 h). Then anhydrous $MgSO_4$ was added and the resulting mixture was diluted with diethyl ether (10 mL) and CH_2Cl_2 (5 mL). The resulting mixture was then filtered through a short pad of silica gel to remove the inorganic salts; eluent: 3×[CH_2Cl_2 (5 mL) + Et_2O (10 mL)]. After evaporation, the mixture was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 20:1) to afford **5i** (365.8 mg, 86%) as an oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.66 (d, J = 8.0 Hz, 2H, ArH), 7.22–7.04 (m, 7H, ArH), 7.03–6.97 (m, 2H, ArH), 5.10 (t, J = 3.6 Hz, 1H, TsNCH), 4.07 (d, J = 18.4 Hz, 1H, one proton of CH_2), 3.55 (d, J = 18.4 Hz, 1H, one proton of CH_2), 2.79–2.66 (m, 4H, CH_2×2), 2.34 (s, 3H, CH_3 of Ts), 2.22–2.09 (m, 1H, one proton of CH_2), 2.05–1.95 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.2, 140.2, 137.2, 135.1, 134.4, 129.5, 129.1, 127.5, 127.4, 126.9, 126.8, 126.5, 125.5, 124.0, 122.5, 121.5, 50.0, 44.2, 27.9, 27.6, 25.7, 21.4; MS (EI) m/z (%) 421 (M^+, 16.23), 265 (100); IR (neat) 2928, 2824, 1597, 1489, 1445, 1338, 1326, 1301, 1249, 1200, 1162, 1091, 1046, 1005 cm^{-1}; HRMS (EI) calcd for $C_{24}H_{23}NO_2S_2$ (M^+): 421.1170; found: 421.1168.

10. Preparation of 2-(3-furanyl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5j



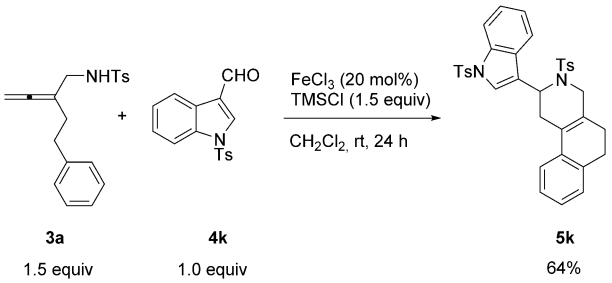
Following **Typical Procedure II**, the reaction of $FeCl_3$ (32.3 mg, 0.20 mmol), **3a** (491.3 mg, 1.50 mmol), **4j** (96.0 mg, 1.00 mmol), TMSCl (165.0 mg, 1.50 mmol), and CH_2Cl_2 (10 mL) afford the desired product **5j** (206.3 mg, 51%)

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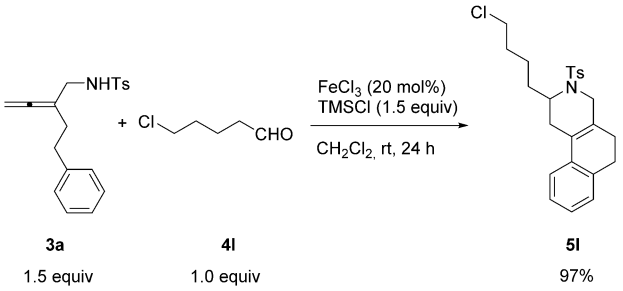
(eluent: petroleum ether/ethyl acetate = 20:1) as an solid; m.p. 150–151 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.69 (d, J = 7.6 Hz, 1H, ArH), 7.26–7.08 (m, 7H, ArH), 7.05 (d, J = 6.8 Hz, 1H, ArH), 6.25 (s, 1H, ArH), 5.40 (d, J = 6.4 Hz, 1H, TsNCH), 4.06 (d, J = 18.4 Hz, 1H, one proton of CH_2), 3.60 (d, J = 18.0 Hz, 1H, one proton of CH_2), 2.84–2.69 (m, 3H, one proton of CH_2 and CH_2), 2.61 (d, J = 16.8 Hz, 1H, one proton of CH_2), 2.36 (s, 3H, CH_3 of Ts), 2.25–2.11 (m, 1H, one proton of CH_2), 2.08–1.98 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.3, 142.9, 140.2, 137.2, 135.0, 134.4, 129.5, 129.0, 127.4, 126.9, 126.8, 126.5, 124.0, 123.6, 121.5, 110.2, 46.9, 44.1, 28.5, 27.7, 25.7, 21.4; MS (EI) m/z (%) 405 (M^+, 17.63), 250 (100); IR (neat) 2927, 2830, 1597, 1490, 1445, 1339, 1304, 1265, 1161, 1094, 1050, 1005 cm^{-1}; anal calcd for $C_{24}H_{23}NO_3S$: C, 71.09; H, 5.72; N, 3.45; found: C, 70.85; H, 5.64; N, 3.51.

11. Preparation of 2-(1-tosyl-1*H*-indol-3-yl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5k



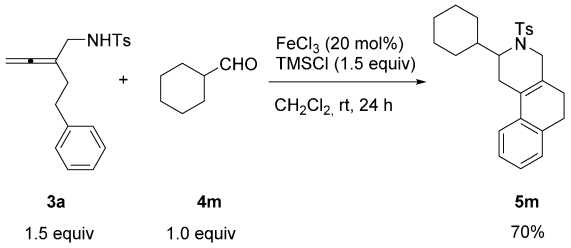
Following **Typical Procedure II**, the reaction of $FeCl_3$ (32.3 mg, 0.20 mmol), **3a** (491.0 mg, 1.50 mmol), **4k** (299.3 mg, 1.00 mmol), TMSCl (165.0 mg, 1.50 mmol), and CH_2Cl_2 (10 mL) afford the impure product **5k** (466.2 mg) (eluent: petroleum ether/ethyl acetate = 10:1) as a solid; recrystallization from CH_2Cl_2 and petroleum ether afforded the desired product **5k** (388.7 mg, 64%) as a solid; m.p. 211–212 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.98–7.91 (m, 2H, ArH), 7.63 (d, J = 8.4 Hz, 2H, Ar-H), 7.56 (d, J = 8.4 Hz, 2H, Ar-H), 7.35–7.24 (m, 2H, Ar-H), 7.20–7.05 (m, 8H, Ar-H), 6.93–6.87 (m, 1H, Ar-H), 5.68 (d, J = 5.2 Hz, 1H, TsNCH), 4.06 (d, J = 19.6 Hz, 1H, one proton of CH_2), 3.35 (d, J = 19.2 Hz, 1H, one proton of CH_2), 2.78–2.70 (m, 2H, CH_2), 2.66–2.51 (m, 2H, CH_2), 2.28 (s, 3H, CH_3 of Ts), 2.26 (s, 3H, CH_3 of Ts), 2.20–2.06 (m, 1H, one proton of CH_2), 2.01–1.91 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 144.9, 143.5, 137.0, 135.1, 134.8, 134.7, 134.1, 129.7, 129.5, 129.4, 129.0, 127.4, 126.8, 126.7, 126.45, 126.43, 125.0, 124.1, 124.0, 123.6, 121.3, 120.8, 120.6, 113.5, 47.1, 44.3, 27.7, 26.2, 25.8, 21.4, 21.3; MS (MAIDI) m/z (%) 631 ([M+Na]$^+$, 100); IR (neat) 2962, 2924, 1597, 1491, 1447, 1370, 1339, 1304, 1262, 1162, 1118, 1093, 1042, 1013 cm^{-1}; anal calcd for $C_{35}H_{32}N_2O_4S_2$: C, 69.05; H, 5.30; N, 4.60; found: C, 68.85; H, 5.36; N, 4.69.

12. Preparation of 2-(4-chlorobutyl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5l



Following **Typical Procedure I**, the reaction of $FeCl_3$ (6.5 mg, 0.040 mmol), **3a** (98.1 mg, 0.30 mmol), **4l** (24.8 mg, 0.20 mmol), TMSCl (32.6 mg, 0.30 mmol), and CH_2Cl_2 (2 mL) afford the desired product **5l** (85.9 mg, 97%) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 20:1:1) as an oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.68 (d, J = 8.4 Hz, 2H, Ar-H), 7.22 (d, J = 8.0 Hz, 2H, Ar-H), 7.19–7.06 (m, 3H, Ar-H), 6.96 (d, J = 6.8 Hz, 1H, Ar-H), 4.25 (q, J = 6.6 Hz, 1H, TsNCH), 4.12 (d, J = 18.8 Hz, 1H, one proton of CH_2), 3.75 (d, J = 18.8 Hz, 1H, one proton of CH_2), 3.47 (t, J = 6.6 Hz, 2H, CH_2), 2.81–2.69 (m, 2H, CH_2), 2.49–2.38 (m, 1H, one proton of CH_2), 2.35 (s, 3H, CH_3 of Ts), 2.25–2.02 (m, 3H, CH_2 and one proton of CH_2), 1.83–1.64 (m, 1H, one proton of CH_2), 1.60–1.34 (m, 4H, 2 × CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.2, 137.6, 135.0, 134.7, 129.6, 128.1, 127.3, 126.7, 126.6, 126.4, 123.6, 121.4, 50.9, 44.7, 43.6, 32.1, 30.5, 27.7, 27.5, 25.6, 23.8, 21.4; MS (EI) m/z (%) 429 (M^+ (^{35}Cl), 2.05), 182 (100); IR (neat) 2930, 2863, 2835, 1598, 1490, 1450, 1384, 1336, 1264, 1156, 1095, 1043, 1012 cm^{-1}; HRMS (EI) calcd for $C_{24}H_{28}NO_2SCl$ (M^+ (^{35}Cl)): 429.1529; found: 429.1533.

13. Preparation of 2-cyclohexyl-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5m



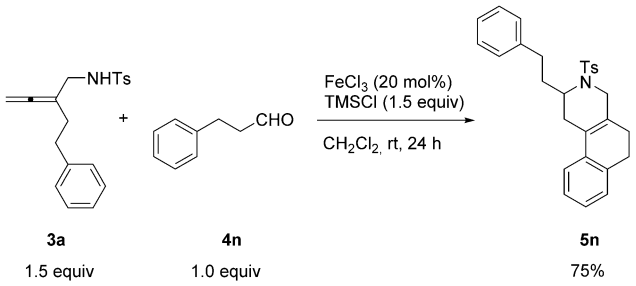
Following **Typical Procedure I**, the reaction of $FeCl_3$ (6.3 mg, 0.040 mmol), **3a** (96.2 mg, 0.30 mmol), **4m** (22.7 mg, 0.20 mmol), TMSCl (32.6 mg, 0.30 mmol), and CH_2Cl_2 (2 mL) afford the desired product **5m** (59.6 mg, 70%) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 20:1:1) as an oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.65 (d, J = 8.1 Hz, 2H, Ar-H), 7.22–7.06 (m, 5H, Ar-H), 6.96 (d, J = 6.9 Hz, 1H, Ar-H), 4.10 (d, J = 19.2 Hz, 1H, one proton of CH_2),

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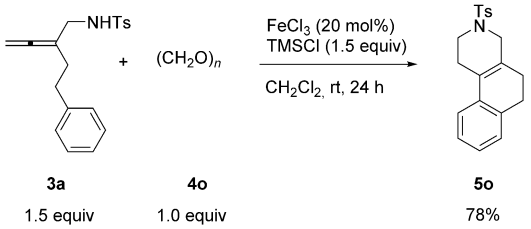
3.88 (dd, J=10.2, 6.0 Hz, 1H, TsNCH), 3.76 (d, J=19.2 Hz, 1H, one proton of CH_2), 2.82–2.65 (m, 2H, CH_2), 2.43 (d, J=15.9 Hz, 1H, one proton of CH_2), 2.33 (s, 3H, CH_3 of Ts), 2.28–1.56 (m, 8H, one proton of CH_2+CH_2+5 protons in cyclohexyl group), 1.53–1.38 (m, 1H, one proton in cyclohexyl group), 1.20–0.87 (m, 5H, 5 protons in cyclohexyl group); ^{13}C NMR (100 MHz, $CDCl_3$) δ 142.9, 138.2, 135.1, 134.8, 129.5, 128.5, 127.3, 126.6, 126.5, 126.4, 123.8, 121.3, 56.4, 44.2, 37.2, 30.9, 30.4, 27.7, 26.2, 26.1, 26.0, 25.6, 23.8, 21.4; MS (EI) m/z (%) 421 (M^+, 1.74), 182 (100); IR (neat) 2960, 2924, 2851, 1599, 1492, 1448, 1338, 1260, 1159, 1092, 1015 cm^{-1}; HRMS (EI) calcd for $C_{26}H_{31}NO_2S$ (M^+): 421.2076; found: 421.2071.

14. Preparation of 2-phenethyl-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5n



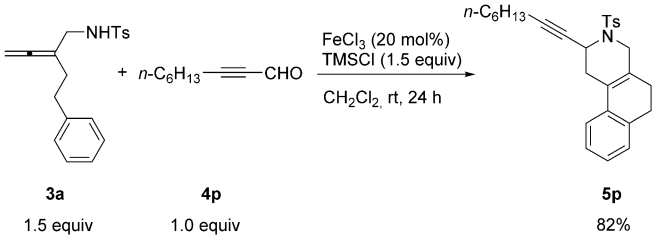
Typical Procedure III: To a dried Schlenk tube were added $FeCl_3$ (32.3 mg, 0.20 mmol)/CH_2Cl_2 (2.5 mL), **3a** (491.3 mg, 1.50 mmol)/CH_2Cl_2 (2.5 mL), **4n** (133.0 mg, 1.00 mmol)/CH_2Cl_2 (2.5 mL), and TMSCl (163.0 mg, 1.50 mmol)/CH_2Cl_2 (2.5 mL) sequentially. The mixture was then stirred at room temperature until the completion of the reaction as monitored by TLC (24 h). The resulting mixture was diluted with diethyl ether (20 mL). Then anhydrous $MgSO_4$ was added and the resulting mixture was then filtered through a short pad of silica gel to remove the inorganic salts (eluent: CH_2Cl_2/Et_2O=1:2, 100 mL). After evaporation, the mixture was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/CH_2Cl_2=40:1:1) to afford **5n** (330.8 mg, 75%) as a solid; m.p. 53–55 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.67 (d, J=8.0 Hz, 2H, Ar-H), 7.29–7.05 (m, 10H, Ar-H), 6.95 (d, J=7.2 Hz, 1H, Ar-H), 4.31 (q, J=6.8 Hz, 1H, TsNCH), 4.16 (d, J=19.2 Hz, 1H, one proton of CH_2), 3.77 (d, J=18.8 Hz, 1H, one proton of CH_2), 2.79–2.59 (m, 4H, 2×CH_2), 2.50–2.39 (m, 1H, one proton of CH_2), 2.34 (s, 3H, CH_3 of Ts), 2.26–2.01 (m, 3H, CH_2 and one proton of CH_2), 1.84–1.62 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.1, 141.4, 137.5, 135.0, 134.7, 129.6, 128.28, 128.27, 128.1, 127.3, 126.8, 126.6, 126.4, 125.8, 123.6, 121.3, 51.0, 43.6, 33.0, 32.7, 27.64, 27.55, 25.6, 21.4; MS (EI) m/z (%) 443 (M^+, 10.27), 182 (100); IR (neat) 3060, 3026, 2926, 2859, 2832, 1599, 1494, 1453, 1385, 1339, 1304, 1289, 1269, 1160, 1097, 1010 cm^{-1}; HRMS (EI) calcd for $C_{30}H_{27}NO_2S$ (M^+): 465.1763; found: 465.1760; anal calcd for $C_{28}H_{29}NO_2S$: C, 75.81; H, 6.59; N, 3.16; found: C, 75.67; H, 6.64; N, 2.97.

15. Preparation of 3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5o



Following **Typical Procedure III**, the reaction of $FeCl_3$ (32.6 mg, 0.20 mmol), **3a** (491.6 mg, 1.50 mmol), **4o** (30.2 mg, 1.00 mmol), TMSCl (162.3 mg, 1.50 mmol), and CH_2Cl_2 (10 mL) afford the impure product **5o** (290.4 mg) (eluent: petroleum ether/diethyl ether/CH_2Cl_2=20:1:1) as a solid; recrystallization from CH_2Cl_2 and petroleum ether afforded the desired product **5o** (269.7 mg, 78%); m.p. 146–147 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.72 (d, J=8.4 Hz, 2H, Ar-H), 7.34 (d, J=8.0 Hz, 2H, Ar-H), 7.20–7.06 (m, 4H, Ar-H), 3.62 (s, 2H, $TsNCH_2$), 3.31 (t, J=5.8 Hz, 2H, CH_2), 2.72 (t, J=7.8 Hz, 2H, CH_2), 2.60–2.53 (m, 2H, CH_2), 2.42 (s, 3H, CH_3 of Ts), 2.12 (t, J=7.6 Hz, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.6, 134.6, 134.4, 132.8, 129.6, 128.6, 127.7, 127.3, 126.6, 126.4, 125.6, 121.6, 48.0, 43.1, 27.6, 25.9, 25.2, 21.5; MS (EI) m/z (%) 339 (M^+, 12.06), 183 (100); IR (neat) 3062, 3028, 2926, 2832, 1597, 1490, 1459, 1452, 1342, 1306, 1166, 1096, 1009 cm^{-1}; anal calcd for $C_{20}H_{21}NO_2S$: C, 70.77; H, 6.24; N, 4.13; found: C, 70.77; H, 6.32; N, 4.04.

16. Preparation of 2-(oct-1-yn-1-yl)-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5p



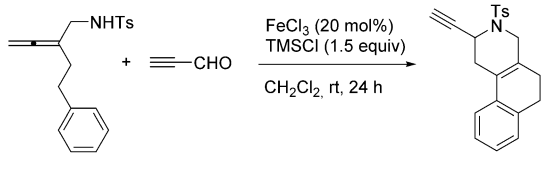
Following **Typical Procedure III**, the reaction of $FeCl_3$ (32.4 mg, 0.20 mmol), **3a** (491.2 mg, 1.50 mmol), **4p** (138.4 mg, 1.00 mmol), TMSCl (164.0 mg, 1.50 mmol), and CH_2Cl_2 (10 mL) afford the desired product **5p** (368.4 mg, 82%) (eluent: petroleum ether/diethyl ether=20:1) as an oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.78 (d, J=8.4 Hz, 2H, Ar-H), 7.30 (d, J=8.0 Hz, 2H, Ar-H), 7.20–7.04 (m, 4H, Ar-H), 5.08 (d, J=4.8 Hz, 1H, TsNCH), 3.97 (d, J=16.4 Hz, 1H, one proton of CH_2), 3.50 (d, J=16.0 Hz, 1H, one proton of CH_2), 2.96 (d, J=17.6 Hz, 1H, one proton of CH_2), 2.74 (dd, J=9.6, 6.0 Hz, 2H, CH_2), 2.53 (d, J=16.0 Hz, 1H, one proton of CH_2), 2.41 (s, 3H, CH_3 of Ts), 2.31–2.16 (m, 1H, one proton of CH_2), 2.12–2.01 (m, 1H, one proton of CH_2), 1.79–1.61 (m, 2H, CH_2), 1.23–0.98 (m,

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8H, 4×CH_2), 0.83 (t, J=7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.3, 134.9, 134.7, 133.9, 129.1, 128.4, 128.3, 127.2, 126.5, 126.4, 123.4, 121.5, 85.3, 74.8, 44.7, 44.0, 32.5, 31.1, 28.2, 28.0, 27.7, 25.8, 22.4, 21.4, 18.2, 14.0; MS (EI) *m/z* (%) 447 (M^+, 17.39), 292 (100); IR (neat) 3063, 3029, 2930, 2857, 2832, 2259, 2223, 1915, 1661, 1598, 1490, 1452, 1442, 1383, 1351, 1306, 1291, 1166, 1126, 1094, 1006 cm^{-1}; HRMS (EI) calcd for $C_{28}H_{33}NO_2S$ (M^+): 447.2232; found: 447.2231.

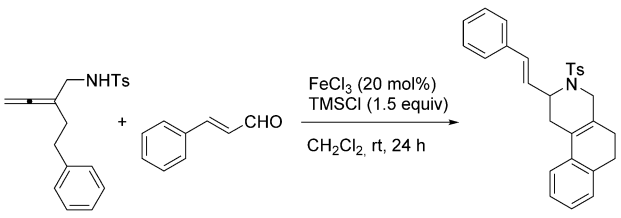
17. Preparation of 2-ethynyl-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline 5q



3a 1.5 equiv | **4q** 1.0 equiv | **5q** 51%

Following **Typical Procedure III**, the reaction of $FeCl_3$ (32.4 mg, 0.20 mmol), **3a** (492.0 mg, 1.50 mmol), **4q** (54.3 mg, 1.00 mmol), TMSCl (164.0 mg, 1.50 mmol), and CH_2Cl_2 (10 mL) afford the desired product **5q** (187.2 mg, 51%) (eluent: petroleum ether/ethyl acetate=30:1) as a solid; m.p. 162–164 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.77 (d, J=8.4 Hz, 2H, Ar-H), 7.30 (d, J=8.0 Hz, 2H, Ar-H), 7.20–7.03 (m, 4H, Ar-H), 5.13 (d, J=6.0 Hz, 1H, TsNCH), 4.00 (d, J=16.4 Hz, 1H, one proton of CH_2), 3.57 (d, J=16.4 Hz, 1H, one proton of CH_2), 2.97 (d, J=14.4 Hz, 1H, one proton of CH_2), 2.79–2.70 (m, 2H, one proton of CH_2 and one proton of CH_2), 2.59 (d, J=16.0 Hz, 1H, one proton of CH_2), 2.40 (s, 3H, CH_3 of Ts), 2.29–2.16 (m, 1H, one proton of CH_2), 2.15–2.05 (m, 1H, one proton of CH_2), 1.84 (d, J=2.0 Hz, 1H, ≡CH); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.7, 134.8, 134.4, 133.8, 129.2, 128.5, 128.2, 127.3, 126.7, 126.4, 123.1, 121.5, 78.7, 72.8, 44.5, 43.5, 31.9, 27.6, 25.8, 21.5; MS (EI) *m/z* (%) 363 (M^+, 11.90), 207 (100); IR (neat) 3286, 3062, 3030, 2926, 2830, 1597, 1490, 1444, 1344, 1326, 1165, 1094, 1052 cm^{-1}; anal calcd for $C_{22}H_{21}NO_2S$: C, 72.70; H, 5.82; N, 3.85; found: C, 72.88; H, 5.88; N, 3.78.

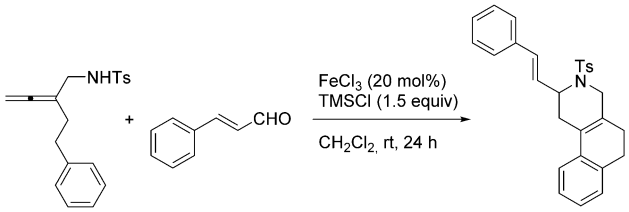
18. Preparation of (*E*)-2-styryl-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline *E*-5r



3a 1.5 equiv | ***E*-4r** 1.0 equiv | ***E*-5r** 82%

Following **Typical Procedure III**, the reaction of $FeCl_3$ (32.5 mg, 0.20 mmol), **3a** (491.5 mg, 1.50 mmol), *E*-**4r** (132.5 mg, 1.0 mmol), TMSCl (163.8 mg, 1.50 mmol), and CH_2Cl_2 (10 mL) afford the desired product *E*-**5r** (362.5 mg, 82%) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2=20:1:1) as an oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.72 (d, J=8.4 Hz, 2H, ArH), 7.28–7.02 (m, 11H, ArH), 6.43 (d, J=16.4 Hz, 1H, =CH), 5.92 (dd, J=16.0, 6.8 Hz, 1H, =CH), 5.02 (t, J=5.6 Hz, 1H, TsNCH), 4.11 (d, J=18.0 Hz, 1H, one proton of CH_2), 3.71 (d, J=18.0 Hz, 1H, one proton of CH_2), 2.85 (d, J=14.8 Hz, 1H, one proton of CH_2), 2.80–2.65 (m, 2H, CH_2), 2.52 (d, J=16.4 Hz, 1H, one proton of CH_2), 2.31 (s, 3H, CH_3 of Ts), 2.29–2.15 (m, 1H, one proton of CH_2), 2.13–2.02 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.3, 136.4, 136.2, 135.0, 134.5, 132.6, 129.5, 128.42, 128.36, 127.7, 127.42, 127.38, 126.7, 126.5, 126.3, 125.5, 123.7, 121.6, 52.9, 44.4, 29.8, 27.7, 25.8, 21.4; MS (EI) *m/z* (%) 441 (M^+, 5.77), 91 (100); IR (neat) 2960, 2926, 2856, 1492, 1452, 1342, 1260, 1163, 1093, 1016 cm^{-1}; HRMS (EI) calcd for $C_{28}H_{27}NO_2S$ (M^+): 441.1763; found: 441.1761.

A large scale synthesis



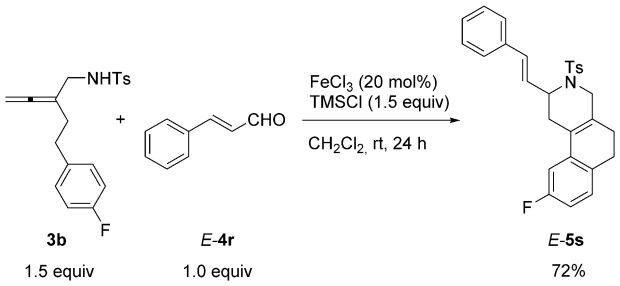
3a 1.5 equiv | ***E*-4r** 1.0 equiv | ***E*-5r** 79%

To a dried Schlenk tube were added $FeCl_3$ (130.1 mg, 0.80 mmol)/CH_2Cl_2 (10 mL), *E*-**4r** (528.6 mg, 4.0 mmol)/CH_2Cl_2 (10 mL), **3a** (1.9651 g, 6.0 mmol)/CH_2Cl_2 (10 mL), and TMSCl (651.5 mg, 6.0 mmol)/CH_2Cl_2 (10 mL) sequentially. The mixture was then stirred at room temperature until the completion of the reaction as monitored by TLC (24 h). The resulting mixture was diluted with diethyl ether (80 mL). Then anhydrous $MgSO_4$ was added and the resulting mixture was then filtered through a short pad of silica gel to remove the inorganic salts (eluent: CH_2Cl_2/Et_2O=1/2, 200 mL). After evaporation, the mixture was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/CH_2Cl_2=20:1:1) to afford *E*-**5r** (1.3977 g, 79%) an oil; δ 7.72 (d, J=8.4 Hz, 2H, ArH), 7.26–7.04 (m, 11H, ArH), 6.43 (d, J=16.0 Hz, 1H, =CH), 5.92 (dd, J=16.0, 6.8 Hz, 1H, =CH), 5.02 (t, J=6.4 Hz, 1H, TsNCH), 4.11 (d, J=18.0 Hz, 1H, one proton of CH_2), 3.71 (d, J=17.6 Hz, 1H, one proton of CH_2), 2.84 (d, J=16.0 Hz, 1H, one proton of CH_2), 2.78–2.68 (m, 2H, CH_2), 2.51 (d, J=16.4 Hz, 1H, one proton of CH_2), 2.30 (s, 3H, CH_3 of Ts), 2.29–2.15 (m, 1H, one proton of CH_2), 2.13–2.01 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.3, 136.4, 136.2, 135.0, 134.5, 132.6, 129.5, 128.4, 128.3, 127.6, 127.4, 127.3, 126.7, 126.5, 126.3, 125.5, 123.7, 121.6, 52.8, 44.4, 29.8, 27.7, 25.8, 21.3;

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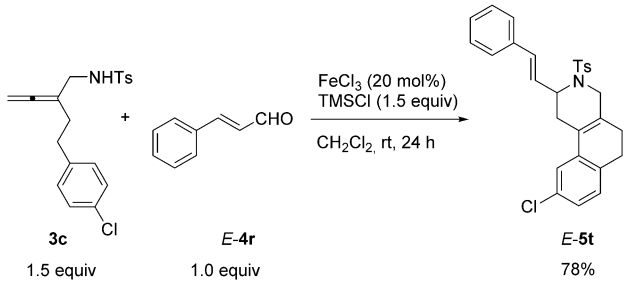
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19. Preparation of (*E*)-9-fluoro2-styryl-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline *E*-5s



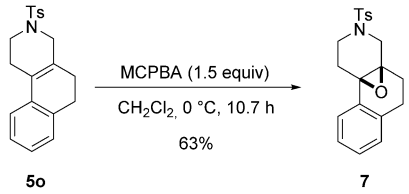
Following **Typical Procedure III**, the reaction of $FeCl_3$ (32.6 mg, 0.20 mmol), **3b** (519.2 mg, 1.50 mmol), *E*-**4r** (132.9 mg, 1.00 mmol), TMSCl (165.0 mg, 1.50 mmol), and CH_2Cl_2 (10 mL) afford the desired product *E*-**5s** (388.4 mg) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 40:1:1) as a solid. Recrystallization from CH_2Cl_2 and petroleum ether afforded the desired product *E*-**5s** (331.6 mg, 72%) as solid; m.p. 118–119 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.72 (d, J = 8.4 Hz, 2H, Ar-H), 7.26–7.15 (m, 5H, Ar-H), 7.12–7.08 (m, 2H, Ar-H), 7.06–7.00 (m, 1H, Ar-H), 6.83–6.73 (m, 2H, Ar-H), 6.41 (d, J = 16.0 Hz, 1H, =CH), 5.89 (dd, J = 16.0, 6.8 Hz, 1H, =CH), 5.01 (t, J = 6.4 Hz, 1H, TsNCH), 4.13 (d, J = 18.0 Hz, 1H, one proton of CH_2), 3.71 (d, J = 18.0 Hz, 1H, one proton of CH_2), 2.84–2.61 (m, 3H, CH_2 + one proton of CH_2), 2.43 (d, J = 16.4 Hz, 1H, one proton of CH_2), 2.30 (s, 3H, CH_3 of Ts), 2.27–2.03 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 161.8 (d, J = 241.4 Hz), 143.3, 136.3 (d, J = 7.6 Hz), 136.2, 136.1, 132.7, 130.3 (d, J = 3.1 Hz), 129.9, 129.4, 128.3, 128.2, 127.7, 127.3, 126.3, 125.1, 123.2 (d, J = 2.3 Hz), 112.7 (d, J = 20.8 Hz), 108.8 (d, J = 22.8 Hz), 52.7, 44.3, 29.7, 26.8, 25.9, 21.3; ^{19}F NMR (376 MHz, $CDCl_3$) −116.8; MS (EI) *m/z* (%) 459 (M^+, 2.87), 91 (100); IR (neat) 3017, 2931, 2837, 1673, 1607, 1582, 1494, 1457, 1439, 1361, 1336, 1327, 1280, 1160, 1136, 1093, 1005 cm^{-1}; anal calcd for $C_{28}H_{26}NO_2SF$: C, 73.18; H, 5.70; N, 3.05; found: C, 72.82; H, 6.03; N, 2.89.

20. Preparation of (*E*)-9-chloro2-styryl-3-tosyl-1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline *E*-5t



Following **Typical Procedure III**, the reaction of $FeCl_3$ (32.7 mg, 0.20 mmol), **3c** (543.2 mg, 1.50 mmol), *E*-**4r** (131.6 mg, 1.00 mmol), TMSCl (165.0 mg, 1.50 mmol), and CH_2Cl_2 (10 mL) afford the desired product *E*-**5t** (370.9 mg, 78%) (eluent: petroleum ether/ethyl acetate/CH_2Cl_2 = 40:1:1) as a solid; m.p. 98–100 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.72 (d, J = 8.4 Hz, 2H, Ar-H), 7.26–7.15 (m, 5H, Ar-H), 7.12–7.06 (m, 3H, Ar-H), 7.04–6.99 (m, 2H, Ar-H), 6.42 (d, J = 16.0 Hz, 1H, =CH), 5.89 (dd, J = 16.2, 6.6 Hz, 1H, =CH), 5.00 (t, J = 6.3 Hz, 1H, TsNCH), 4.12 (d, J = 18.0 Hz, 1H, one proton of CH_2), 3.71 (d, J = 17.6 Hz, 1H, one proton of CH_2), 2.83–2.62 (m, 3H, CH_2 + one proton of CH_2), 2.44 (d, J = 16.8 Hz, 1H, one proton of CH_2), 2.31 (s, 3H, CH_3 of Ts), 2.28–2.14 (m, 1H, one proton of CH_2), 2.13–2.02 (m, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.4, 136.2, 136.13, 136.09, 133.2, 132.7, 132.1, 129.9, 129.5, 128.4, 128.3, 127.7, 127.4, 126.33, 126.30, 125.2, 123.1, 121.8, 52.7, 44.3, 29.6, 27.0, 25.7, 21.3; MS (EI) *m/z* (%)477 (M^+(^{37}Cl), 3.77), 475 (M^+(^{35}Cl), 7.45), 320 (100); IR (neat) 3059, 3026, 2924, 2890, 2833, 1596, 1487, 1448, 1346, 1164, 1094, 1008 cm^{-1}; anal calcd for $C_{28}H_{26}NO_2SCl$: C, 70.65; H, 5.51; N, 2.94; found: C, 70.37; H, 5.58; N, 2.76.

21. Epoxidation of 1,2,3,4,5,6-hexahydrobenzo[*f*]isoquinoline derivative 5o



To a dried Schlenk tube were added **5o** (339.6 mg, 1.0 mmol)/CH_2Cl_2 (5.0 mL). After cooling in ice bath, MCPBA (85% purity, 304.5 mg, 1.5 mmol)/CH_2Cl_2 (5 mL) was added, and then the resulting solution was warmed up to room temperature. The mixture was stirred at room temperature until completion of the reaction as monitored by TLC (10.7 h). After evaporation, Et_2O (40 mL) was added, and the organic layer was washed sequentially with sat. aq. $Na_2S_2O_3$, sat. aq. $NaHCO_3$ and satd. aq. NaCl sequentially. Then anhydrous Na_2SO_4 was added and the resulting mixture was then filtered. After evaporation, the mixture was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 15:1) to afford 7 (224.6 mg, 63%) as a solid; m.p. 157–158 °C (petroleum ether/CH_2Cl_2); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.67 (d, J = 8.0 Hz, 2H, Ar-H), 7.43–7.36 (m, 1H, Ar-H), 7.33 (d, J = 8.0 Hz, 2H, Ar-H), 7.28–7.20 (m, 2H, Ar-H), 7.11–7.05 (m, 1H, Ar-H), 3.87 (d, J = 13.2 Hz, 1H, one proton of CH_2), 3.77–3.65 (m, 1H, one proton of CH_2), 3.11 (d, J = 13.2 Hz, 1H, one proton of CH_2), 2.81 (td, J = 14.5, 6.4 Hz, 1H, one proton of CH_2), 2.70–2.50 (m, 4H, 2 × CH_2), 2.43 (s, 3H, CH_3 of Ts), 2.18 (dd, J = 14.4, 5.8 Hz, 1H, one proton of CH_2), 1.91 (td, J = 14.0, 5.2 Hz, 1H, one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.6, 136.8, 133.1, 132.6, 129.6, 128.4, 128.2, 127.4, 126.3, 126.1, 62.5, 57.8, 46.9, 39.6, 26.5, 26.2, 25.4, 21.3; MS (EI) *m/z* (%) 355 (M^+, 3.15), 200 (100); IR (neat) 2962, 2922, 2851, 1597, 1494, 1457, 1440, 1340, 1305, 1261, 1161, 1095, 1009 cm^{-1}; anal calcd for $C_{20}H_{21}NO_3S$: C, 67.58; H, 5.95; N, 3.94; found: C, 67.42; H, 5.93; N, 3.91.

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